

RAFAEL DE CAMILLO MASSON

Computer Engineer - 23 years old

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PROFESSIONAL SUMMARY

Computer Engineer focused on **Artificial Intelligence** and **Data Science**. Skilled in applying **Machine Learning** and **Deep Learning** to diverse challenges, ranging from cybersecurity and socioeconomic analysis to satellite imagery. Experienced in the full data lifecycle, including preprocessing, model development, and interactive visualization. Background includes practical experience in **Computer Vision** and cloud-based training workflows.

HARD SKILLS

- **Languages:** Python, C/C++, SQL.
 - **Data Science & AI:** Machine Learning, Deep Learning, Computer Vision, Data Analysis, Data Structures and Algorithms.
 - **Libraries & Frameworks:** PyTorch, TensorFlow, OpenCV, NumPy, Pandas, Matplotlib, Scikit-learn, Plotly, Seaborn.
 - **Tools:** Power BI, Kubeflow, Streamlit, Jupyter Notebook, Git.
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PROFESSIONAL EXPERIENCE

SiDi - Intelligence & Innovation Center | *Computer Vision Intern* | 2024 – 2025

- Developed and enhanced segmentation and denoising models for both synthetic and real images.
 - Managed and preprocessed large-scale image datasets for neural network training.
 - Performed data analysis and validation to ensure training data quality.
 - Utilized **PyTorch** and **OpenCV** to implement computer vision solutions.
 - Leveraged **Kubeflow** for cloud-based training workflows.
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EDUCATION

Federal University of São Carlos (UFSCar)

- Bachelor's Degree in Computer Engineering | 2020 – 2025
 - **Undergraduate Thesis:** "Identification of Water Tanks and Swimming Pools as Potential Dengue Breeding Sites Using Satellite Images".
 - **Undergraduate Research:** "Data Analysis and Machine Learning for Intrusion Detection".
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TECHNICAL PROJECTS

Global Development Analysis (World Bank Data) | *Data Analyst / Developer*

- Analyzed global socioeconomic indicators (WDI dataset) using **Pandas**, **NumPy**, and **Scikit-learn**.
- Segmented countries into development tiers using **Unsupervised Learning** (**K-Means** & **HDBSCAN** clustering).
- Engineered **Supervised ML** classification (95.45% accuracy) and regression models ($R^2 = 0.82$) with robust cross-validation.
- Developed an interactive **Streamlit** dashboard with a live predictive simulator and **Plotly** visualizations.
- **Roadmap:** Implementing **Time-Series Analysis** to track development evolution from 2000 to the present.

Patient Readmission Risk Pipeline & ROI Dashboard | *Analytics Engineer / MLE*

- Built an end-to-end machine learning and data engineering pipeline in **Python** using **SQLite**, **XGBoost**, and **Streamlit**.
- Normalized medical transactional logs into **SQLite** tables and leveraged **SQL** for in-database feature engineering.
- Optimized an **XGBoost** model via 5-Fold Stratified Cross-Validation, achieving 81.19% accuracy and 83.08% AUC.
- Designed a business optimization framework that simulates clinical preventive programs, achieving a projected 198% ROI (\$18.36M net savings).
- Deployed an interactive **Streamlit** dashboard featuring a real-time clinical risk scorer and dynamic ROI simulators.

COURSES & CERTIFICATES

- **Machine Learning Scientist in Python** – DataCamp (2025)
- **Data Analyst in Power BI** – DataCamp (2025)
- **Associate Data Analyst in SQL** – DataCamp (2025)
- **Associate Data Scientist in Python** – DataCamp (2025)
- **Data Analyst in Python** – DataCamp (2025)
- **Python Data Fundamentals** – DataCamp (2025)

LANGUAGES

- **Portuguese:** Native proficiency.
- **English:** Full professional proficiency.
- **Spanish:** Limited working proficiency.